

DANIEL RAMOS

SOFTWARE ENGINEER | XR & INTERACTIVE SYSTEMS

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EXPERIENCES

Apprentice Backend Developer (AI) | Sky Portugal

Jul 2025 - Aug 2025

- Designed and implemented an anomaly detection pipeline to identify piracy patterns in Peacock streaming logs, delivering a solution adopted by the piracy analysis team in production.
- Evaluated and benchmarked multiple unsupervised ML approaches (DBSCAN, Isolation Forest, Autoencoders), selecting the optimal model for the use case.
- Engineered a batching system capable of processing 10 million log entries in under 3 minutes, paired with a filtering interface.

Tech Stack: Python, ElasticSearch, ScikitLearn, Tensorflow, NumPy, Streamlit

VR Development Intern | Realizasom

Feb 2024 - Aug 2024

- Sole developer of a deployed MR museum experience for Meta Quest 3 ([Ciclo da Truta de Góis](#)), built end-to-end in Unity/C# with OpenXR – graded 18/20.
- Built a spatial AR system overlaying interactive points of interest onto a physical map, with 360° videos and images of regional landmarks.
- Optimized for accessibility and usability in public exhibition contexts for a non-technical audience.

Tech Stack: Unity, C#, OpenXR, Fork

Software Engineer | O Seu Vizinho

Jun 2023 - Jul 2023

- Co-developed a full-stack caregiver platform for elderly assistance (rides, groceries, pharmacy) across backend, frontend, and database in 2 weeks.
- Pitched at Nova impACT! Challenges, drawing post-competition interest from a jury member.

Tech Stack: PHP, Laravel, MySQL, XAMPP, HTML, CSS, Bootstrap

PROJECTS

Enhancing XR Experiences Through Multi-Device Asymmetric Collaboration

Currently developing

Thesis Project

- Designed and implemented an asymmetric networked MR framework enabling cooperation between Meta Quest 3 (VR) and Android tablet (AR) clients.
- Solved cross-device spatial alignment (Meta XR SDK + AR Foundation) and built a custom Go server with WebSockets/UDP for real-time state synchronization.
- Demonstrated at Expo FCT: a tablet user spawned ingredients for a VR player to collect in their physical environment.

Tech Stack: Unity, C#, Meta XR SDK, AR Foundation, Go, WebSockets, UDP

GeoDescobrimentos Website

Personal Project

- Developed an interactive website that allows users to explore routes, captains and other information related to the portuguese discoveries.
- Built front-end, backend and database optimized to support geospatial data including coordinates and polygons for maps.

Tech Stack: PHP, QGIS, XAMPP, HTML, Javascript, Turf.js, CSS, Bootstrap

EDUCATION

NOVA School of Science and Technology

Semptember 2021 - Present

MSc in Computer Science and Engineering | Current Grade: 17

Major: Interactive Multimodal Systems

BSc in Computer Science and Engineering | Final Grade: 16

SKILLS

Languages: C#, Python, Go, Java, C, Javascript

Backend & Systems: WebSockets, UDP, REST APIs, ElasticSearch

Tools & Infrastructure: Azure, Kubernetes, Git, Linux

ML & Data: Scikit-learn, TensorFlow, NumPy, Pandas